SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

PROJECT NAME: REPLATE

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TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Database Management System
2	Backend Development Framework
3	Mapping and Location Services API

SOFTWARE TOOLS FOR TASK 1: Database Management System

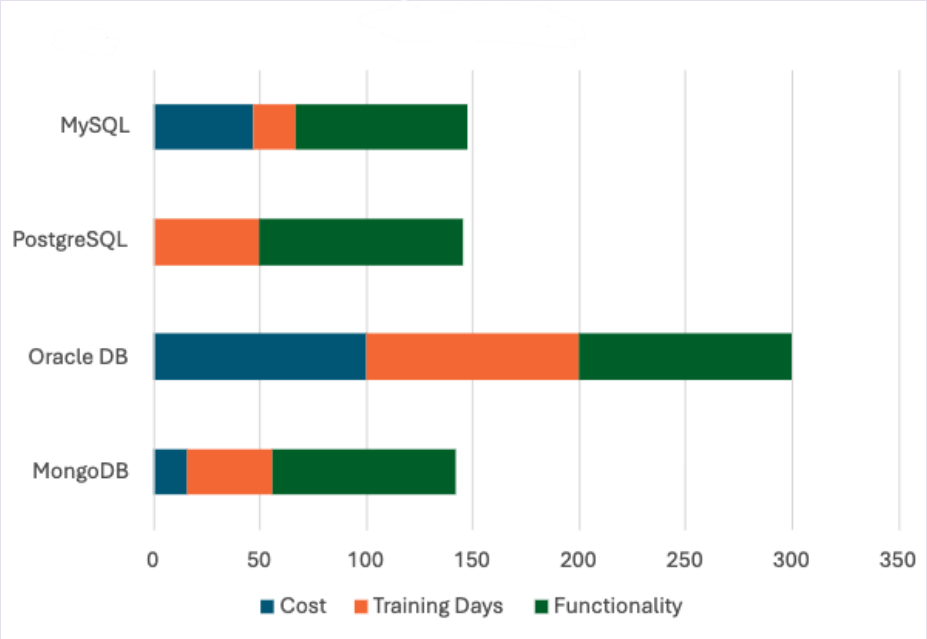
Tool Cost/Training/Functionality Data

Tool	MySQL	PostgreSQL	Oracle DB	MongoDB
Cost	$2250/y	$0	$4800/y	$784/y
Training Days	2	5	10	4
Functionality	80	95	99	85

Normalized Cost/Training/Functionality Data

Tool	MySQL	PostgreSQL	Oracle DB	MongoDB
Cost	46.9	0	**100**	16.3
Training Days	20	50	**100**	40
Functionality	80.8	96	**100**	85.9

Normalized Tool Graph



Which tool has been selected? Why?

For the RePlate project, we have selected PostgreSQL as our Database Management System (DBMS). As seen in the normalized tool graph, PostgreSQL provides the optimal balance by offering exceptionally high functionality (96) while keeping the software license cost completely free (0).

Although Oracle DB provides the highest raw functionality (100), it maximizes both the cost (100) and training days (100) indexes, making it highly inefficient and overly complex for our project's scope and budget. Conversely, MySQL and MongoDB have significantly lower functionality scores (80.8 and 85.9, respectively), which may not fully support our advanced system requirements. RePlate heavily relies on complex relational data—connecting users, businesses, reservations, and municipal distribution centers—as well as highly dependent location-based queries to display surplus food on an interactive map. PostgreSQL natively supports advanced spatial data processing through its PostGIS extension, making it the perfect fit to handle our map-based distance filtering and strict 25-meter GPS accuracy requirement. Furthermore, its moderate training requirement (50) ensures that our development team can adapt to it without causing severe delays, making it the most cost-effective and powerful solution for our system.

SOFTWARE TOOLS FOR TASK 2: Backend Development Framework

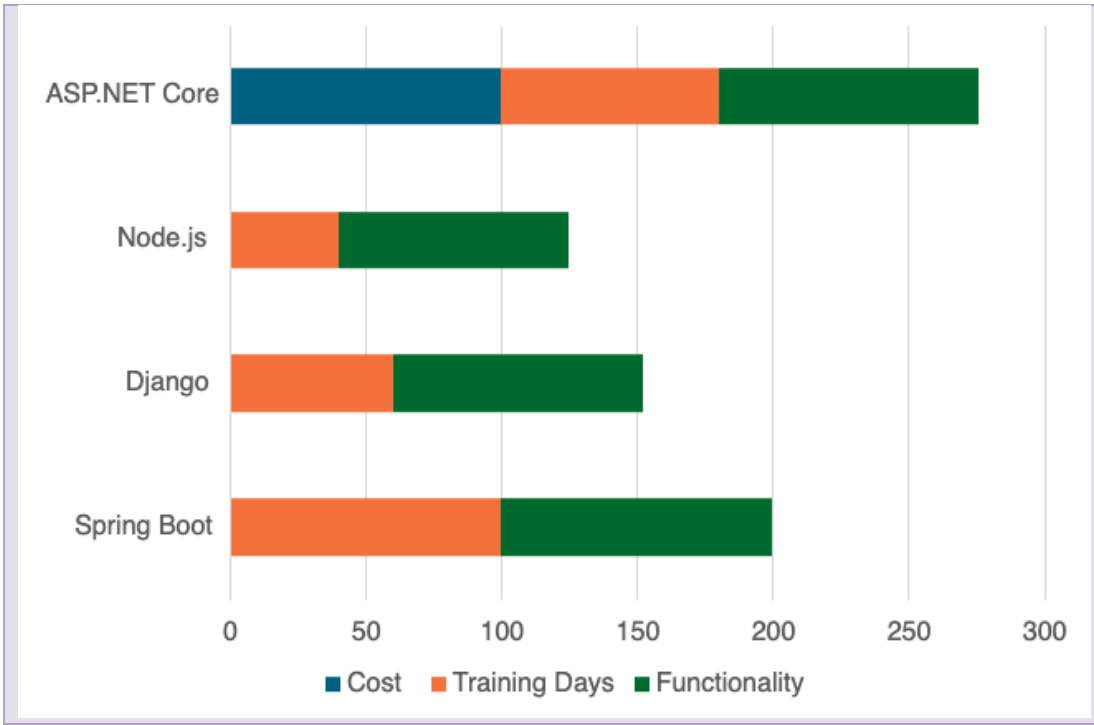
Tool Cost/Training/Functionality Data

Tool	ASP.NETCore	Node.js Exp	Django Py	Spring Boot jv
Cost	$3000/y	$0	$0	$0
Training Days	8	4	6	10
Functionality	96	85	92	100

Normalized Cost/Training/Functionality Data

Tool	ASP.NETCore	Node.js Exp	Django Py	Spring Boot jv
Cost	100	0	**0**	0
Training Days	80	40	**60**	100
Functionality	96	85	**92**	100

Normalized Tool Graph



Which tool has been selected? Why?

For the RePlate project, we have selected Spring Boot (Java) as our Backend Development Framework. When analyzing our framework options, while Node.js and Django offer faster training periods (40 and 60, respectively), Spring Boot provides the absolute maximum Functionality (100) with a completely free open-source license ($0), avoiding the massive $3000 licensing costs associated with enterprise environments like ASP.NET Core.

RePlate is a highly complex system requiring strict data integrity between users, businesses, and municipalities to execute our automated two-stage redistribution model (FR-12). Spring Boot is a robust, Java-based enterprise framework that integrates flawlessly with our chosen PostgreSQL database to manage these complex relational schemas securely. Furthermore, this architecture is highly adaptable to cross-platform development, allowing our team to seamlessly code and test the backend whether they are operating on Windows or macOS environments. We also have a strict non-functional requirement (NFR-7) stating the system must support 10,000 concurrent users with response times under 3 seconds. Spring Boot's powerful multithreading capabilities and scalable nature guarantee this high-performance handling. Although it maximizes our training days index (100), investing this time is crucial to building a secure, fail-proof backend engine.

SOFTWARE TOOLS FOR TASK 3: Mapping and Location Services API

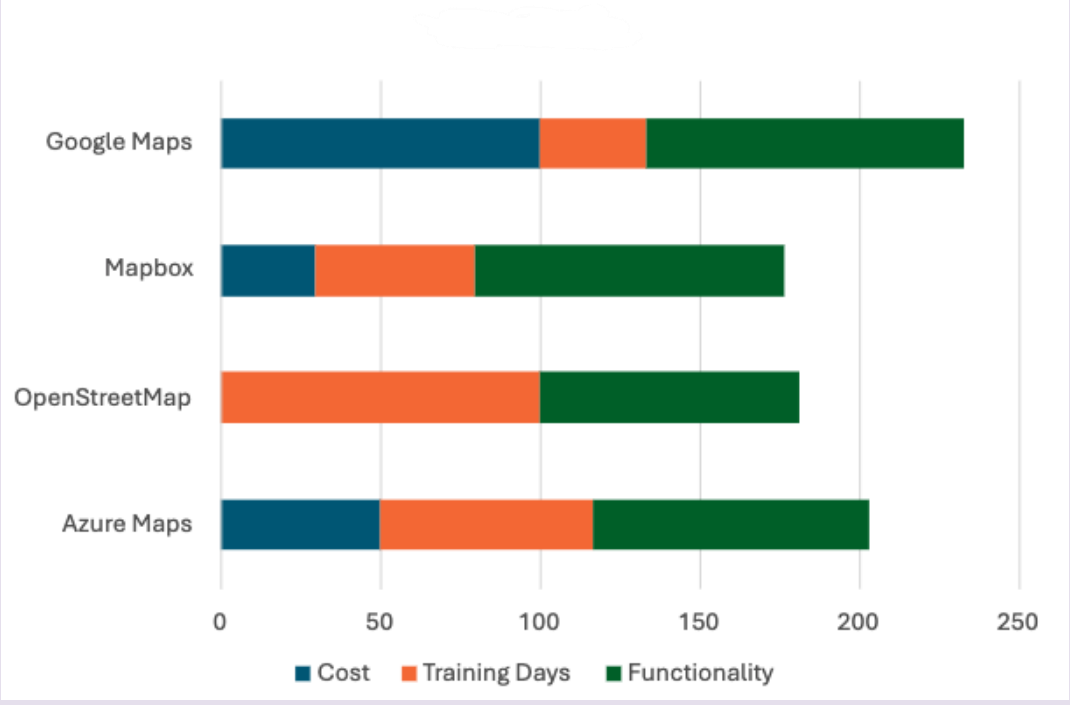
Tool Cost/Training/Functionality Data

Tool	Google Maps	Mapbox	OpenStreetM	Azure Maps
Cost	$2000/y	$600/y	$0	$1000/y
Training Days	2	3	6	4
Functionality	98	95	80	85

Normalized Cost/Training/Functionality Data

Tool	Google Maps	Mapbox	OpenStreetM	Azure Maps
Cost	100	30	**0**	50
Training Days	33.3	50	**100**	66.7
Functionality	100	96.9	**81.6**	86.7

Normalized Tool Graph



Which tool has been selected? Why?

For the RePlate project, we selected Google Maps as our Mapping and Location Services API. According to our normalized data, while Mapbox offers a lower cost, Google Maps provides the absolute highest Functionality (100) and requires the lowest Training Days (33.3) for our development team.

The primary reason for this selection is its unparalleled global ubiquity and user familiarity. As the industry standard, Google Maps is natively integrated into almost all Android devices and widely used on iOS, which perfectly aligns with our requirement (NFR-9) to support diverse mobile platforms. Because everyday users are already intimately familiar with the Google Maps interface, it significantly increases the likelihood of satisfying our strict usability requirement (NFR-6), where at least 90% of users must complete a product reservation in under 3 minutes without external assistance. Furthermore, its industry-leading spatial data accuracy guarantees we will easily meet our 25-meter maximum GPS error margin requirement (NFR-11). Although it possesses the highest cost index (100), the massive reduction in team training time and the superior, universally accepted user experience make Google Maps the most reliable and powerful mapping tool for RePlate.